

Lab 3: Direct RNA Sequencing of Lung and Liver Tissue

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Sample Sheet

RNA quality for both lung and liver samples was assessed using the Agilent Fragment Analyzer system, using the High Sensitivity Total RNA kit (DNF-472). The resulting profiles allow estimation of RNA concentration, 28S/18S ratios, and the RNA Quality Number (RQN).

Key metrics for each tissue:

- Liver (e.g. Sample Liver 14):
 - RQN: 7.2
 - 28S/18S Ratio: 1.4
 - Total RNA Concentration: 5.34 ng/μL
- Lung (e.g. Sample Lung 09):
 - RQN: 7.7
 - 28S/18S Ratio: 1.0
 - Total RNA Concentration: 7.09 ng/μL

RQN values above 7.0 indicate high-quality RNA suitable for direct RNA sequencing. Although the 28S/18S ratio in lung tissue was slightly lower than in liver, both samples exceeded the minimum quality threshold typically required for native RNA protocols.

Pre-Lab Questions

The 28S/18S ratio for lung is lower than liver, but the RQN is higher. What does this suggest about the relationship between these two RNA quality metrics?

A: _____

Why is it important to avoid RNA degradation when preparing libraries for direct RNA sequencing?

A: _____

What are some common sources of RNA degradation during extraction, and how might you mitigate them in this lab?

A: _____

Review the protocol. Why do we use fluorometric quantification methods in Step 9?

A: _____

Library Preparation Workflow

Day 1: Reverse Transcription and Adapter Ligation

1. Transfer 1 µg total RNA to a 0.2 mL PCR tube. Adjust volume to 8 µL with nuclease-free water.
2. In the same tube, add: 3 µL NEBNext Quick Ligation Buffer, 0.5 µL RNA CS (optional), 1 µL Murine RNase Inhibitor, 1 µL RT Adapter, 1.5 µL T4 DNA Ligase. Mix and spin.
3. Incubate 10 min at room temp.
4. Prepare RT master mix: 13 µL water, 2 µL 10 mM dNTPs, 8 µL 5× Induro RT Buffer. Add to ligated sample, then add 2 µL Induro RT enzyme.
5. Incubate: 60°C 30 min → 70°C 10 min → 4°C hold.
6. Add 72 µL RNAClean XP beads. Rotate 5 min, wash ×2 with 70% ethanol. Elute in 23 µL water.

Q: What is happening to the RNA at each temperature?

A: _____

Day 2: Final Adapter Ligation and Clean-Up

7. Mix: 23 µL RT-RNA, 8 µL Quick Ligation Buffer, 6 µL RNA Ligation Adapter, 3 µL T4 DNA Ligase. Incubate 10 min.
8. Add 16 µL RNAClean XP beads, rotate 5 min, wash ×2 with 150 µL WSB.
9. Elute in 33 µL REB. Quantify 1 µL with Qubit dsDNA HS assay.

Q: Why is Qubit used here instead of Nanodrop?

A: _____

Loading the PromethION Flow Cell

10. Thaw and mix: 30 µL RNA Flush Tether, 1170 µL Flow Cell Flush. Load 500 µL for priming.
11. Prepare library mix: 100 µL SB, 68 µL LIS, 32 µL RNA Library. Total: 200 µL.
12. Load 200 µL into flow cell. Install light shield.

Q: Why is it critical to maintain continuous buffer coverage over the nanopore array during flow cell loading?

A: _____

Post-Sequencing: Bioinformatics Notes

Software: MinKNOW [on-machine] or Dorado [off-machine]

Basecalling: Dorado [<https://github.com/nanoporetech/dorado>]

Goals: Transcript quantification, RNA modification detection (m6A, Ψ, m5C, inosine), tissue-specific epitranscriptomic profiles

Post-Lab Questions

- Q: How could you detect ligation inefficiencies or poor RNA quality from the sequencing output?

A: _____

- Q: Suggest one way to normalize data between samples if input RNA quantity or yield varied.

A: _____

- Q: What can we learn from direct RNA sequencing in the context of an epitranscriptomic atlas?

A: _____

- Q: Why might liver and lung tissues have different RNA modification profiles?

A: _____

- Q: What are the benefits and disadvantages to using native RNA instead of cDNA?

A: _____